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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 1

Limits, Continuity and Differentiability

1.1. LIMITS

If $f(x)$ tends to L as x tends to a , we say that the limit of $f(x)$ as x tends to a is L . This

is abbreviated by $\lim_{x \rightarrow a} f(x) = L$

$\lim_{x \rightarrow a} f(x)$ is the value that $f(x)$ approaches as x approaches a , which means that for

each real $\epsilon > 0$ there exists a real number $\delta > 0$ such that for all x within

$$0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$$



Definitions; (i) $\lim_{x \rightarrow a^+} f(x)$ is the limit of $f(x)$ as x tends to a through values

greater than a . [Right hand limit *RHL*]

(ii) $\lim_{x \rightarrow a^-} f(x)$ is the limit of $f(x)$ as x tends to a through values

less than a . [Left hand limit *LHL*]

Theorem

Suppose that $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} g(x) = L_2$

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = L_1 + L_2$



$$2. \lim_{x \rightarrow a} (f(x)g(x)) = L_1 L_2$$

$$3. \lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \left(\frac{L_1}{L_2} \right)$$

$$4. \lim_{x \rightarrow a} \lambda(f(x)) = \lambda L_1$$

Example . Evaluate $\lim_{x \rightarrow 2} (3x^3 - x^2 + 2x + 5)$

Solution: $\lim_{x \rightarrow 2} (3x^3 - x^2 + 2x + 5) = 3 \lim_{x \rightarrow 2} x^3 - \lim_{x \rightarrow 2} x^2 +$

$$2 \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 5$$

$$= 3(2)^3 - (2)^2 + 2(2) + 5 = 29$$



Example  Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x + 1}$

Solution: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x + 1} = \frac{0}{2} = 0$ or

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x+1)} = x - 1 = 0$$

Example  Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

Solution: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)} = x + 1 = 2$



Example . Evaluate $\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x}$

Solution:
$$\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x} = \lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{(1+\sqrt{x})(1-\sqrt{x})} = \frac{1}{1+\sqrt{x}} = \frac{1}{2}$$

Example .
$$\lim_{x \rightarrow \infty} \frac{5x^2-3x+2}{10x^2-x+100}$$

Solution:
$$\lim_{x \rightarrow \infty} \frac{5x^2-3x+2}{10x^2-x+100} = \lim_{x \rightarrow \infty} \frac{5-\frac{3}{x}+\frac{2}{x^2}}{10-\frac{1}{x}+\frac{100}{x^2}} = \frac{5}{10} = \frac{1}{2}$$

Example .
$$\lim_{x \rightarrow \infty} \sqrt{1+\frac{1}{x}} - \sqrt{\frac{1}{x}}$$

Solution:
$$\lim_{x \rightarrow \infty} \sqrt{1+\frac{1}{x}} - \sqrt{\frac{1}{x}} =$$



$$\lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x} - \frac{1}{x}}{\sqrt{1 + \frac{1}{x}} + \sqrt{\frac{1}{x}}} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + \sqrt{\frac{1}{x}}} = 1$$

□

De L'Hopital's Rule

Suppose $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$, but $g'(x) \neq 0$

Then
$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(x)}{g'(x)}$$

Examples

1.
$$\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x} = \lim_{x \rightarrow 1} \frac{0 - \frac{1}{2}x^{-1/2}}{-1} = \frac{1}{2}$$

2.
$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{6x} = \lim_{x \rightarrow 0} \frac{\cos x}{6} = \frac{1}{6}$$

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$$3. \lim_{x \rightarrow 0} \frac{x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{1}{2 \cos 2x} = \frac{1}{2}$$

$$4. \lim_{x \rightarrow 0} \frac{\sin 7x}{4x} = \lim_{x \rightarrow 0} \frac{7 \cos 7x}{4} = \frac{7}{4}$$

$$5. \lim_{y \rightarrow \frac{\pi}{2}} \left(\frac{\cos y}{\frac{\pi}{2} - y} \right) = \lim_{y \rightarrow \frac{\pi}{2}} \left(\frac{-\sin y}{-1} \right) = 1$$

$$6. \lim_{x \rightarrow 0} \left(\frac{x \sin x}{1 - \cos x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x + x \cos x}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\cos x + \cos x - x \sin x}{\cos x} \right) = \frac{2}{1} = 2$$

Example  $\lim_{x \rightarrow 0} x^x$

Solution: Let $y = x^x$ then, $\ln y = x \ln x$





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let $x = e^t$ then $\ln y = e^t \ln e^t = te^t$

as $x \rightarrow 0, t \rightarrow 0 \therefore te^t \rightarrow 0$ and as $\ln y \rightarrow 0, y \rightarrow 1$

Hence $\lim_{x \rightarrow 0} x^x = 1$

EXERCISE 1.  $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

More Revision Questions

Evaluate the following limits

$$(1) \lim_{y \rightarrow 0} \left(\frac{y^2 + 5y + 9}{y + 2} \right)$$

$$(2) \lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right)$$

$$(3) \lim_{x \rightarrow 1} \left(\frac{x^2 - x - 2}{x^2 - 1} \right)$$

$$(4) \lim_{x \rightarrow 0} \left(\frac{x^3}{\sin 2x} \right)$$





$$(5) \lim_{x \rightarrow 0} \left(\frac{\sin 7x}{5x} \right)$$

$$(7) \lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right) \left(\frac{5x^2 - 1}{x^2} \right)$$

$$(9) \lim_{\theta \rightarrow 0} \left(\frac{\sin^2 \theta}{\theta^2} \right)$$

$$(11) \lim_{x \rightarrow \infty} \left(\frac{5x+1}{10+2x} \right)$$

$$(13) \lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x} \right)$$

$$(6) \lim_{x \rightarrow \infty} \left(\frac{3x^2 - 6x}{4x - 8} \right)$$

$$(8) \lim_{x \rightarrow \infty} \left(\frac{7x-28}{x^3} \right)$$

$$(10) \lim_{\theta \rightarrow 0} \left(\frac{1 - \cos \theta}{\theta} \right)$$

$$(12) \lim_{x \rightarrow \infty} \left(\frac{x+1}{x^2} \right)$$

$$(14) \lim_{x \rightarrow \infty} \left(\frac{5}{1+x} \right)$$

1.2. CONTINUITY

Definition Suppose that the function f is defined in a neighborhood of a . We say that f is continuous at a provided that



the $\lim_{x \rightarrow a} f(x)$ exists and moreover that the value of this limit is $f(a)$. In other words f is continuous at a provided that $\lim_{x \rightarrow a} f(x) = f(a)$.

For a function f to be continuous at the point a it must satisfy the following three conditions:

1. The function f must be defined at a (so that $f(a)$ exists)
2. The limit of $f(x)$ as x approaches a must exist
3. The numbers in condition 1. and 2. must be equal i.e.

$$\lim_{x \rightarrow a} f(x) = f(a)$$

If any one of these conditions is not satisfied, then f is not continuous at a . If the function f is not continuous at a , the we



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say that f is discontinuous or that a is a discontinuous of f .

Example . $f(x) = \frac{1}{x-2}$ for $x \neq 2$

Solution: $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{1}{x-2} = -\infty$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{1}{x-2} = \infty$$

Therefore $f(x)$ is discontinuous at $x = 2$ (Infinite discontinuity at $x = 2$)

□

Example . $g(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$

Solution: Its left hand limit and right hand limit at $x = 0$ are unequal i.e.



$$\lim_{x \rightarrow 0^-} g(x) = -1 \text{ and } \lim_{x \rightarrow 0^+} g(x) = 1$$

Thus $g(x)$ has no limit as $x \rightarrow 0$. Hence the function g is not continuous at $x = 0$, it has what might be called a finite jump discontinuity there.

EXERCISE 2.  $h(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$



Solutions to Exercises

Exercise 1. Let $y = \left(1 + \frac{1}{x}\right)^x$, then $\ln y = x \ln \left(1 + \frac{1}{x}\right)$

Let $z = \frac{1}{x}$, then $x = \frac{1}{z} \quad \therefore \ln y = \frac{1}{z} \ln(1 + z)$

as $x \rightarrow \infty$, $z \rightarrow 0$

$$\lim_{z \rightarrow 0} \frac{\ln(1+z)}{z} = \lim_{z \rightarrow 0} \frac{1/(1+z)}{1} = 1$$

as $x \rightarrow \infty$, $z \rightarrow 0$ and $\ln y \rightarrow 1 \implies y \rightarrow e$

$$\therefore \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

Exercise 1



Exercise 2. $\lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$ whereas $h(0) = 0$.

$\Rightarrow$ The limit and the value of h at $x = 0$ are not equal. Thus the function h is not continuous there.

Exercise 2